

Backed Engineer Assignment (v5)

Goal

Build a small web application that generates a **report** of game transaction data in **MySQL 8** within a date range. The report must support **sorting and filtering per column**.

Tech Requirements

- **Java 8+**
 - **Spring (MVC)**
 - **JSP** (view layer)
 - **Stripes** (Optional)
 - **Hibernate / JPA**
 - **MySQL 8**
 - **Maven 3.8**
 - **Docker** (Optional)
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Data Model

You will be provided with a script that can create a table and populate the data. You will need to setup a MySQL database and use the script to create a table and populate the data. You may create other indexes or tables if you need. In this case, please provide the SQL script to create the indexes and/or tables.

Web UI (Spring MVC + JSP)

Create a web page with:

A) Search Form

Inputs:

- **Start date/time**
- **End date/time**
- Optional **Account ID**
- Submit button: **Generate Report**

Validation:

- Start <= End

- Date range required

B) Report Table

Display all records in the range with columns:

- `id`
- `account_id`
- `datetime`
- `tran_type`
- `platform_tran_id`
- `game_tran_id`
- `game_id`
- `amount` (sum of all the AMOUNT_* columns)
- `balance` (sum of all the BALANCE_* columns)

C) Sorting (Required)

- Allow sorting by clicking column headers.
- Must support ascending/descending.

D) Filtering Per Column (Required)

Provide basic filters for at least:

- `platform_tran_id`
- `game_tran_id`
- `account_id`
- `game_id`
- `tran_type`

E) Pagination

- Page size: 25 or 50
- Show total count

Submission Requirements

Include:

1. Source code

After you're done, push your project to a public Git repo on one of the following remote Git

repository services:

- a. GitHub
- b. GitLab (specifically [GitLab.com](https://gitlab.com) and not your own instance)
- c. BitBucket

Submit the repo link to: eddie.kim@bet99.com

2. README.md

Create a Markdown readme in the repo's root with information on the project. Your target audience is a newly-hired developer who will work on the service. What do they need to know to (a) get the system compiled and running; and (b) any information they need about the server and client designs that will help them work on the project.

Optional Enhancements (Bonus)

- Export to CSV
- Summary section: totals by **type** (bet sum, win sum), net = win - bet
- Unit tests for service/repository
- Docker compose for MySQL + app

Review

We will do the following:

1. Review the Git repo.
2. Follow the readme and build the server.
3. Run the app and test function(s).
4. We may ask you to share your screen and see your environment setup.